FROM RAW GOVERNMENT DATA TO MEANINGFUL INSIGHTS.

USING PUBLIC DATA TO TRANSFORM NUMBERS INTO INSIGHTS

CRIME AGAINST WOMEN IN INDIA

A DATA ANALYTICS STUDY (2020)

Working within data constraints: making the most of what is available

Crimes against women remain a critical socio-economic issue in India. Understanding trends across crime categories helps policymakers, researchers, and citizens evaluate the effectiveness of laws and social interventions.

This project analyzes India's crime data against women for the last two years, using official government datasets, with a focus on crime-category-wise trends rather than state-wise distribution.

Cruelty by Husband or his relatives >100000

Introduction

In a data-driven world, statistics are only as good as the insights they provide. This project serves as a comprehensive Efficiency Audit of the 2020 National Crime Records Bureau (NCRB) dataset. The objective was to move beyond the simple reporting of crime numbers and instead focus on the Investigation Lifecycle—measuring how effectively cases are processed, chargesheeted, and disposed of by the police.

By analyzing the "Justice Funnel," this project identifies critical bottlenecks where cases remain pending for years, providing a data-backed foundation for policy and resource allocation.

Objective of the Study

- To analyze trends in crimes against women over the last two years
- To compare crime categories and identify dominant patterns
- To demonstrate a beginner-friendly yet technically sound data analysis workflow
- To create a replicable analytical guide for students and early researchers

DATA SOURCE

NCRB (PRIMARY SOURCE)

NATIONAL CRIME RECORDS BUREAU

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Phase 1:The Excel Foundation (Data Engineering)

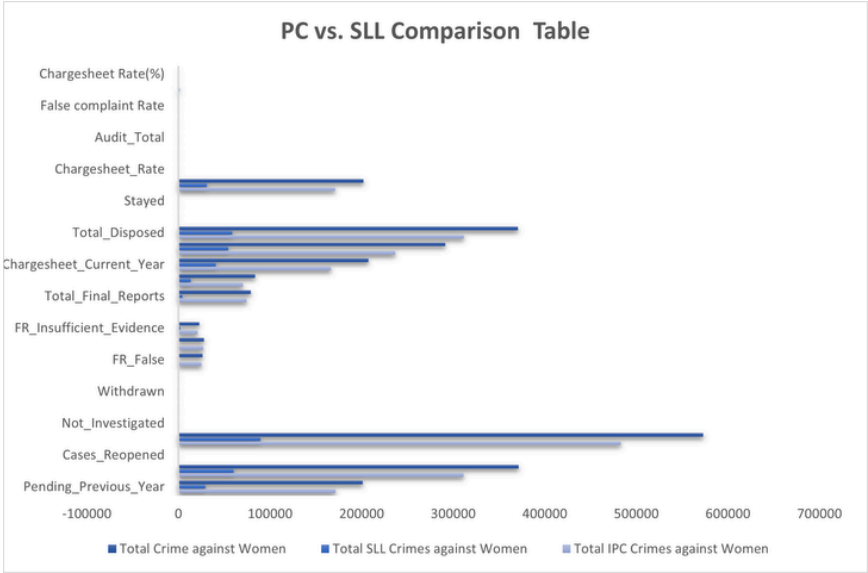
The Need & Importance

Raw government data is often structured for human reading, not machine processing. It contains merged cells, summary rows, and inconsistent formatting. Excel was used as the "Sanitation Layer" to ensure data integrity before any advanced modeling began.

PROCESS & STEPS

1.Data Scrubbing: Identified and removed summary rows (Grand Totals) to prevent "double-counting" in the final analysis.

2.Type Conversion: Standardized columns from text to numeric format using the "Text to Columns" feature, ensuring mathematical operations could be performed.



3.Variable Formatting: Renamed complex column headers into structured, "Snake_Case" variables (e.g., Chargesheet_Rate) for seamless integration with Python scripts.

4.Initial Validation: Used Conditional Formatting to flag outliers—specifically looking for crime categories where "Pending Cases" significantly outweighed "Total Investigations."

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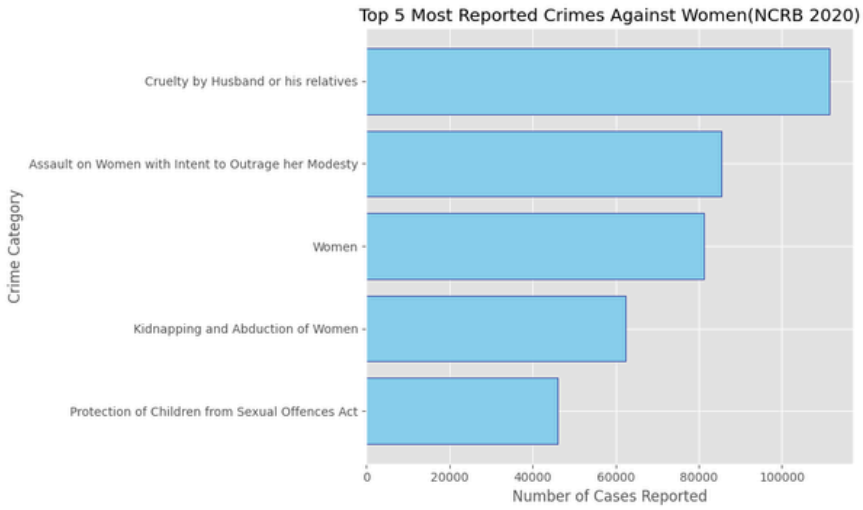
Phase 2: The Python Audit (Scalability & Validation)

The Need & Importance

Excel has limits when dealing with logic-heavy verification. Python was introduced to perform an Automated Audit. The goal was to ensure that the numbers provided by the NCRB were mathematically sound and to generate high-resolution, reproducible visualizations that could be used in professional publications.

PROCESS & STEPS

1Library Integration: Leveraged the Pandas library for high-speed data manipulation and Matplotlib/Seaborn for exploratory plotting.



2.Algorithmic Verification: Wrote a logic check to verify the internal consistency of the data:

Logic: Total\ Disposed == Chargesheets + Final\ Reports + Abated\ Cases.

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3.Exploratory Data Analysis (EDA): Filtered the top 5–10 most prevalent crime categories to provide a focused "Heat Map" of the legal landscape.
4.Exporting Insights: Generated PNG visualizations programmatically, ensuring the analysis is reproducible with any future yearly dataset (e.g., NCRB 2021/22).

Phase 3: Power BI (The Executive Storyteller)

The Need & Importance

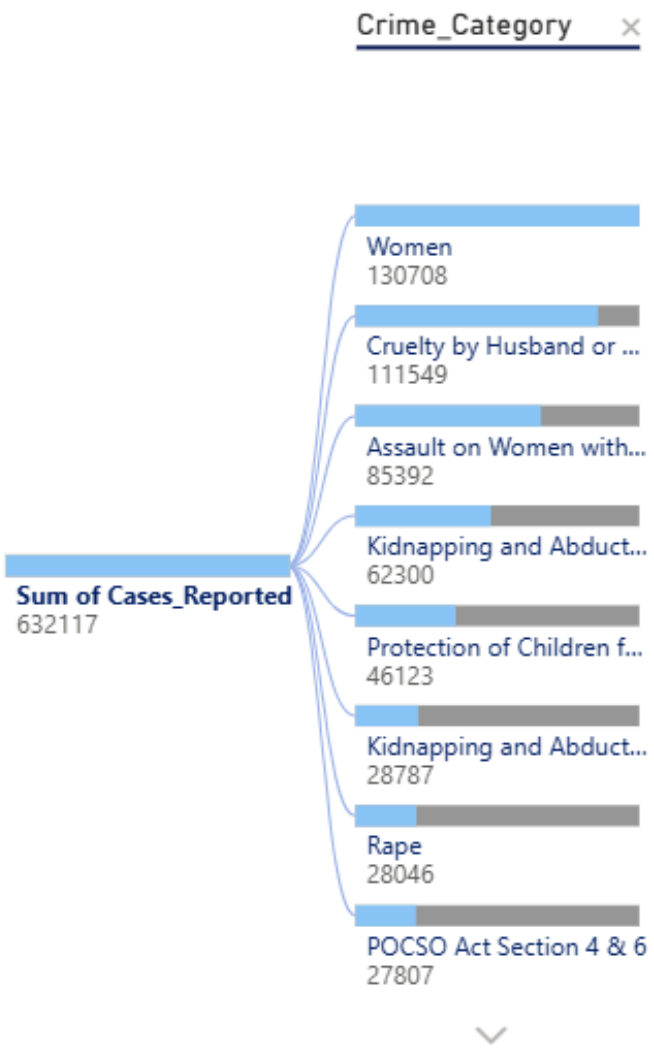
Executives and stakeholders don't want to read code or look at spreadsheets; they need actionable insights. Power BI was used to transform the static data into an interactive Operational Dashboard.

PROCESS & STEPS

1. **Data Modeling: Created a Star Schema by separating Crime Categories into a Dimension table.**

2. **Advanced DAX: Wrote measures for Chargesheet Rate % and National Average to create performance benchmarks.**

3. **AI Visuals: Integrated a Decomposition Tree to allow users to "drill down" into a specific crime and see exactly where the backlog is happening.**



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A Three-Tiered Approach (Excel + Python + Power BI)

The true value of this project lies in the synergy of the tools used. Each tool was selected for its specific strength in the analyst's workflow:

- **Excel (The Foundation): Used for Data Ingestion and Cleaning. It provided the first line of defense against messy data, ensuring that the "garbage in, garbage out" rule was avoided.**
- **Python (The Engine): Used for Validation and Automation. It proved that the analysis was mathematically robust and allowed for complex filtering that would be unstable in a basic spreadsheet.**
- **Power BI (The Storyteller): Used for Executive Visualization. It took the static numbers from Phase 1 and 2 and turned them into a dynamic environment where a stakeholder can "ask questions" of the data through filters, bookmarks, and decomposition trees**

From an analyst's lens, this project demonstrates a shift from Descriptive Analytics (what happened) to Diagnostic Analytics (why there is a backlog). By using the "Consumer Impact Index" logic from my telecom thesis and applying it as a "Justice Efficiency Index" here, I have created a cross-domain methodology that can be applied to any sector—from market monopolies to national security.

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